SYSTEM INFORMATION

MVIL Portal product summary

Motor Vehicles Insurance Limited, Papua New Guinea. One web portal for vehicle owners, MVIL staff and roadside officers. It covers CTP insurance, vehicle registration, safety stickers, driver licences and online payments.

1. Purpose

The MVIL Portal replaces paper and counter-only processes. Records are kept in one place, owners can pay and check expiry dates online, and officers can confirm on the spot whether a vehicle and driver are compliant. It works in any web browser and can be installed on Android phones like an app.

2. Key features

Vehicle owners (members)

- Personal portal with every vehicle owned
- CTP insurance, registration and safety sticker expiry at a glance
- Online payment for services, with PDF and email receipts
- QR codes for vehicles and licences
- Renewal and ownership history, documents and notices
- Expiry reminders by email

MVIL administrators

- Individual and organisation owners, with their vehicles
- Register or renew CTP insurance, registration and safety stickers together for a chosen vehicle
- Drivers and licences, with photos and licence history
- Services, fees and registration prices by make and model
- Staff invitations and roles
- Search by owner name or number plate

Roadside officers

- Scan QR codes on stickers and licences
- Number plate recognition from a photo
- Driver face matching against licence photos (95% confidence by default)
- Instant compliant or not compliant result
- Public verification page for any QR code

Root account (system control)

- Payment gateways with separate Sandbox and Live keys, plus a test payment
- Face matching settings, including an optional AWS engine
- ID requirements: make NID or SevisPass mandatory

- PNG Geo: Province, District, LLG, Ward and Location
- Scheduled jobs and email reminders
- Full audit trail with search, filters and date range

Security and accountability: Role-based access (member, agent, officer, admin, root), secure single sign-on, secrets that are never shown in full, and an audit log of every sign-in and staff action that cannot be edited.

3. Membership onboarding and member services

Vehicle owners and drivers can join the portal online on their own, with no visit to an MVIL office needed. Once registered, they become MVIL members and can use the self-service features below.

1. Membership onboarding

The member signs in securely (Google, email or other supported methods), then completes a short membership form: full name, mobile number, home and postal address with province and district, and optional NID or SevisPass numbers. Root can make either ID mandatory.

2. Claim a driver licence

The member enters their licence number and uploads supporting documents, such as a photo of the licence card and an ID. The claim goes to MVIL for review.

3. Claim vehicle ownership

The member enters a number plate and uploads proof of ownership, such as registration papers or a bill of sale. MVIL checks the documents against its records.

4. Review and decision

MVIL admins and root review each claim with the member's profile and documents side by side, mark it under review, then approve or reject it with a note. A claim cannot be approved until documents are uploaded. The member gets an email at each step: acknowledgement, a reminder if documents are missing, and the final decision.

5. Everyday self-service

Members can pay for services online, download PDF receipts, keep a personal My documents folder (upload, view and delete files), track claim progress and read notices. SevisPass eWallet claiming is shown as coming soon.

4. Real-time payments for MVIL admin

Every online payment is visible to MVIL admins and root in real time, the moment it is made. There is no overnight batch or manual upload. Staff can filter by status (paid, pending, failed, cancelled) and by Live or Sandbox, search by payment reference or payer email, open the full record of any payment, download it as a PDF, and export any date range as a spreadsheet (CSV) for finance or ERP reconciliation. Every export is recorded in the audit trail.

5. Expiry reminders for members

Vehicle registration renewal

Reminds the vehicle owner before the registration expires for each vehicle they own.

Safety sticker

Reminds the vehicle owner before the safety sticker expires, so the vehicle stays roadworthy.

Driver licence

Reminds the driver before their licence expires.

Reminders are sent automatically 14, 7, 3 and 1 days before the expiry date, and again on the day itself. They arrive by email and appear in the member's portal notices, with a link to renew online. Root can switch reminders on or off and see when they last ran.

6. Integration with ERP and other systems

The portal is designed to work alongside MVIL's existing systems, such as ERP, finance and legacy registers, without custom code for each one.

Configurable external data sources

The root account connects any system with a standard web API (REST/JSON) from a settings screen. There are six ready-made connection points: vehicles, owners, registrations, safety stickers, drivers and driver licences.

Common sign-in methods

It supports API key or username and password, a custom header name, and an adjustable timeout. Secrets are stored securely and only ever shown masked.

Flexible lookup paths

Each connection uses a path template, for example /vehicles/{plate}, so it adapts to the ERP's own address format.

Built-in connection test

Root can test each connection and preview the response before switching it on. The last test result is recorded.

Bulk data import

Standards-based building blocks

Reference data can be loaded from spreadsheet (CSV) files. All PNG geographic areas, about 45,000 places, were imported this way.

Payments use Stripe and PayPal, face matching can use Amazon Web Services, and sign-in uses OpenID Connect. Any of these can be swapped or extended.

7. Technologies used

Platform and hosting	Hercules app builder and Hercules Cloud (managed hosting, deployment and scaling)
Web front end	React 19, TypeScript, Vite, Tailwind CSS 4, shadcn UI
Backend and database	Convex: real-time database and server functions with indexed search
Sign-in and security	Hercules Auth (OpenID Connect single sign-on) with role-based access
Payments	Stripe (cards) and PayPal, each with Sandbox and Live modes
Face matching	face-api in the browser, with Amazon Rekognition as an optional extra engine
AI reading	Hercules AI Gateway for reading number plates and stickers from photos
QR codes	qrcode (creating codes) and jsQR (scanning with the camera)
Documents and email	jsPDF for receipts, Hercules Email for receipts, notices and reminders
Mobile	Installable web app (PWA) for Android and desktop
Quality	Vitest automated tests, TypeScript strict checks, ESLint

8. Modules yet to be implemented

SevisPass integration Planned

Live identity checks against the PNG Government's SevisPass digital ID. SevisPass numbers are already recorded and can be made mandatory. Verification currently shows as "coming soon" and is simulated in Sandbox.

Transfer of Ownership (live) Planned

The full seller-to-buyer flow already works as a root-only Sandbox simulation: identity check, new owner acceptance, fee payment and document handover. It goes live once SevisPass is integrated.

BSP payment gateway Planned

Card and bank payments through Bank South Pacific, charged in Kina. It will be added to the existing gateway settings, with its own Sandbox and Live keys.

Kundu payment gateway Planned

Mobile and wallet payments through Kundu Pay for owners without cards. It will appear as another option at checkout.

Further enhancements under consideration Under consideration

CTP insurance status in roadside compliance checks, CTP expiry reminders, and an inbound API so ERP systems can push updates and receive payment records automatically.

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MOTOR VEHICLES INSURANCE

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